

XINGE GUO

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RESEARCH INTERESTS

Computer vision and machine learning for medical image analysis, with a focus on **Representation Learning**, **Image Segmentation**, **Clinical Decision Making** and **Treatment Planning**.

EDUCATION

Duke University, Durham, NC, USA	Aug 2025 – Present
<i>M.S. in Biomedical Engineering · Advisor: Prof. Sina Farsiu (VIP Lab)</i> <i>GPA: 3.81 / 4.00</i>	
• Fellowships: Research Fellowship for BME MS/MEng Students (Fall 2025 & Spring 2026); Merit-based Scholarship (Fall 2025)	
Shenzhen University, Shenzhen, China	Sep 2021 – Jun 2025
<i>B.Eng. in Biomedical Engineering, Class of Excellence</i> <i>GPA: 3.91 / 4.50</i>	
• Honors: Top Innovative Talent Scholarship (Dec 2022, Dec 2023, Dec 2024); Full Scholarship for Class of Excellence (Sep 2021)	

PUBLICATIONS

- Guo X**, Xiao F, Zhang D, Chen Y, Zhang R, Li X, Wang T, He C, Farsiu S. "Can Frozen Hyperspherical Features Guide the Selection of Pseudo Masks?" **arXiv preprint, 2026**, arXiv:2609.30080.
- Guo X**, Wang Y, Shu L, Liu Y, Xu M. "ConPro: Contrast Projection Pretraining for Label-Efficient Vessel Segmentation in DSA Sequences." **arXiv preprint, 2026**, arXiv:2609.30043.
- Xiao F, Hu P, Xu L, **Guo X**, Qin G, Shen Y, Fang C, Zhang R, He C, Farsiu S. "Beyond Ground-Truth: Leveraging Image Quality Priors for Real-World Image Restoration." **IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026**.
- He C, Zhang D, Xiao F, **Guo X**, Feng J, Wang Y, Farsiu S. "Toward Multimodal Understanding in Low-Level Vision: The Rise of Vision-Language Models." Preprints 2026, 202606.1854. Under review at **ACM Computing Surveys**.
- Dai J, **Guo X**, Zhang H, et al. "Cone-beam CT Landmark Detection for Measuring Basal Bone Width: A Retrospective Validation Study." **BMC Oral Health, 2024**, 24(1): 1091. (Co-First Author)

RESEARCH EXPERIENCE

Xu Lab, Computational Biology Department, Carnegie Mellon University	Jun 2026 – Present
<i>Research Intern · Advisor: Prof. Min Xu</i> <i>Pittsburgh, PA, USA</i>	
<i>Project 1 — ConPro: Contrast Projection Pretraining for DSA Vessel Segmentation (arXiv 2026)</i>	
• Self-supervised pretraining that predicts each pixel's contrast projection (normalized drop below its temporal median) in DSA sequences, as a label-efficient initialization for vessel segmentation.	
<i>Project 2 — Mechanistic Interpretability of Medical Vision Transformers (Ongoing)</i>	
• Sparse feature decomposition showing that medical ViTs rely on non-pathological but causally dominant (NPC) features , which explains their poor localization of pathology.	
Vision and Image Processing (VIP) Laboratory, Duke University	Sep 2025 – Present
<i>Graduate Researcher · Advisor: Prof. Sina Farsiu</i> <i>Durham, NC, USA</i>	
<i>Project 1 — SphereTrust: Frozen Hyperspherical Features for Pseudo-Mask Selection (arXiv 2026)</i>	
• Training-free selection of SAM pseudo masks via the hypersphere of frozen DINOv2 features .	
<i>Project 2 — Leveraging Image Quality Priors for Real-World Image Restoration (CVPR 2026)</i>	
• Replaces pixel-level ground-truth supervision with a learned image quality prior for real-world restoration.	

Project 3 — Segmentation Difficulty Benchmark via VLM (Ongoing)

- A **VLM-based agent** that queries multiple segmentation models to predict per-image segmentation difficulty and likely failure regions before deployment.

Department of Orthopaedic Surgery, Duke University School of Medicine

Dec 2025 – Present

Research Collaborator · Advisor: Dr. Brian C. Lau, MD

Durham, NC, USA

Project — Glenoid and Humeral Bone Loss Measurement on 3D CT in Shoulder Instability (Ongoing)

- **3D CT segmentation and quantitative measurement pipeline** for glenoid and humeral bone loss, developed with orthopaedic surgeons.

Medical Ultrasound Engineering Lab, Shenzhen University

Mar 2024 – Apr 2025

Undergraduate Researcher · Advisor: Prof. Chien Ting Chin

Shenzhen, China

Project — Left Ventricle Landmark Detection (Graduation Thesis)

- A **lightweight end-to-end network** that reformulates left ventricle landmark detection.

Dept. of Radiology & Nuclear Medicine, Erasmus MC

Jul 2024 – Sep 2024

Research Intern · Mentor: Dr. Xianjing Liu

Rotterdam, the Netherlands

Project — GWAS on AI-Derived Phenotypes from Structural Brain MRI in UK Biobank

- A **3D convolutional autoencoder** that learns imaging phenotypes from UK Biobank brain MRI for genome-wide association analysis.

Medical AI Laboratory, School of Biomedical Engineering, Shenzhen University

Aug 2022 – Oct 2024

Undergraduate Researcher · Advisor: Prof. Bingsheng Huang

Shenzhen, China

Project — CBCT Landmark Detection for Basal Bone Width Measurement (BMC Oral Health 2024)

- A **U-Net-based model** for orthodontic landmark detection on CBCT images.

SELECTED COURSEWORK

- **Duke (Graduate):** AI Image Assessment (Independent Study with Prof. Farsiu, A+), Signal Processing and Applied Math (A), Ultrasound Imaging (A), Digital Image Processing (Fall 2026, in progress)
- **Shenzhen U (Undergraduate):** Principles of Medical Imaging (A+), Medical Ultrasound Technology (A), Medical Signal Processing (A), Signals and Systems (A), Digital Electronics (A+), Probability and Statistics (A), Linear Algebra (A+), Calculus (A+), Python Programming (A)

TECHNICAL SKILLS

- **Programming:** Python (primary), MATLAB, C/C++, VHDL, Assembly
- **Languages:** English (Fluent, IELTS 7.5), Mandarin (Native), Cantonese (Native)

OTHER

- **Piano Performance:** Associate Diploma (ALCM), London College of Music, University of West London (Apr 2022)